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This is a pre print version of the following article:

Original Citation:

Availability:

This version is available <http://hdl.handle.net/2318/1885145> since 2023-04-29T09:45:47Z

Publisher:

Springer

Published version:

DOI:10.1007/978-3-031-25233-4_13

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DOI: 10.13140/RG.2.2.17064.08965/1

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Agile as a Vehicle for Values: A Value Sensitive Design Toolkit

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Abstract

The ethics of technology has primarily focused on what values are and how they can be embedded in technologies through design. In this context, some work has been done to show the efficacy of a number of design approaches. However, existing studies have not clearly pointed out the ways which design-for-values approaches can be used by design team managers to properly organize and use technologies in practice. This chapter attempts to fill this gap by discussing the value sensitive design (VSD) approach as a useful means of co-designing technologies as a toolkit for existing workflow management, in this case Agile. It will be demonstrated that, VSD shows promise as a way of democratically designing technologies as well as fostering democratic technology policy innovation.

Acknowledgements

We would like to thank Batya Friedman, David Hendry, and other participants at the VSD Quarterly Seminar on 8 April 2021 for providing helpful comments and suggestions on the initial draft of this work. All remaining errors are the authors' alone. The views expressed in this chapter do not necessarily reflect those of the Institute for Ethics and Emerging Technologies or of Ethical Intelligence Associates, Limited.

Introduction

Since at least the 1980s, the philosophy and ethics of technology have been contending that technology and society influence each other (Winner, 2003). In recent years, nation states have been paying attention to the effects of this phenomenon. There are regional attentions to responsible innovation, like the EU Horizon 2020 projects, and international ones, such as with the United Nations Sustainable Development Goals (SDGs). What underlies this global orientation towards responsible innovation of new technologies is the notion that technologies are fundamentally systemic: they interact with social forces and hence cannot be reduced to mere neutral instruments.

The value sensitive design (VSD) approach was developed in the early 1990's essentially, as a means to reflect human values in technology design, rather than arbitrarily referring to them *ex post facto*, or ignoring them completely. For over two decades, VSD has been demonstrated to be uniquely capable of being adopted across a wide range of design domains such as online browsers (Millett et al., 2001), urban simulation (Davis et al., 2006), care robotics (Umbrello et al., 2021; van Wynsberghe, 2013), artificial intelligence (Umbrello, 2019), energy systems (Mouter et al., 2018), and manufacturing technologies (Longo et al., 2020). Despite the extensive scholarship on the topic, as well as the systemic nature of VSD, existing studies have not addressed the issue of how and why VSD should be implemented/adopted at managerial levels within design domains, and not only by dispersed groups of interested engineers. Of course, engineers and engineering teams are often hierarchical and part of larger organizational structures which introduce policies that often govern the working principles of the design team. If VSD seamlessly integrates into the design domain – a fundamental conceptual precept of the approach – it also needs to be applied to the field of management.

This chapter aims to address this critical gap by exploring not only the nature of VSD itself as a design approach capable not only of being applied to the design of technologies themselves (i.e., the artefacts) but also to the responsible designing of the context of use in which it is adopted (i.e., the design domain). Our discussion is divided into the following sections. §1 looks at systems thinking and systems engineering which globally forms the ontological foundation of much of the current trend in technology. §2 briefly outlines the VSD approach as an implementation of systems engineering, and shows how it can satisfy many of the conceptual requirements of systems engineering. §3 shows how VSD is attractive not only to designers and engineers themselves, but also to engineering management which is a direct stakeholder as well as a subject of change. The change in engineering management affects the whole team. We will illustrate the applicability of VSD to engineering management by showing how it fits within the Agile methodology, a widely adopted methodology for project management. §4 discusses how engineering management can begin to think about implementing a VSD toolkit in their Agile process as well as how to increase its symbioses through an explicit orientation towards systems thinking. §5 presents our conclusions.

Systems Thinking and Systems Engineering

There are multiple reasons why systems thinking and systems engineering needs to be introduced in this discussion. Firstly, systems thinking in and of itself characterises the various levels of systems. It does not only look at the system *per se* (the artefact) but also the system in which it forms a part (i.e., the organization). Secondly, VSD is fundamentally a systems engineering approach. Because VSD is fundamentally predicated on the *interactional stance* of technology that we mentioned in the introduction, it examines systems, rather than artefacts in isolation, or the organization that has only ancillary relation to design. As such, the focal points of VSD, like systems thinking, are the plurality of actors, institutions, technologies, and their design histories. Hence, for a responsible innovation to take place, these various connected elements need to be given explicit and primary attention in thinking about design.

Broadly, systems thinking is the interdisciplinary study of organised and complex systems (Whitchurch & Constantine, 2009). The nodes, or parts that form a system, are both covariant as well as co-constitutive of each other, and are thus dynamic in their relationship and complexity. A system's environment and context of use both support and constrain its function, the latter of which is teleological in the sense that its intended use is determined by its operation (Adams et al., 2014). This complexity allows systems to be described as greater than the sum of their parts. The covariance of parts creates behaviours that may not be produced by any of the individual parts. This *synergy* of constituent parts, as well as the emergent behaviour which it generates, specifies a goal of systems thinking: the behaviour mapping of patterns to help predict such behaviours in different environments of use (Haken, 2013).

Systems engineering then takes this more ontological understanding of systems as the theoretical basis for its application in engineering and design. It takes a similarly interdisciplinary approach to the understanding, design, management, and deployment of engineered systems to ensure optimised equifinality over their lifecycles (Thomé, 1993). In order to achieve such equifinality, a system is mapped to determine how the parts that co-constitute it work synergistically and thus provide predictable emergent behaviours in different environments of use. One can already begin to see how such a framing of engineering can be useful when we talk about technologies like artificial intelligence based on machine learning, which learn and adapt based on many (different) environmental inputs.

This interdisciplinary systems of engineering employs many related human-centred areas of research, such as risk analysis, organisational studies, and project management alongside technical studies such as mechanical, electrical, software, and industrial engineering, among others. By doing this, it takes the discipline of engineering itself as a system, to be managed which leads to the conception of the technical artefacts design as *systems-within-systems*. The purpose of this approach is to map and design the way managerial and organizational structures in which these systems are developed both support and constrain certain design decisions.

Value Sensitive Design (VSD)

For almost three decades, value sensitive design has been a popular approach to designing technologies *for* human values. Rather than relegating them to *ex post facto* or *ad hoc* additions

to systems, VSD is espoused as a principled approach to technology design that makes human values the goals of the design. Pioneered by Batya Friedman and her colleagues at the University of Washington in the early 1990's, VSD has since become one of the leading design approaches in the field of engineering ethics and responsible research and innovation (Winkler & Spiekermann, 2018). Although its various applications is outside the scope of this chapter, it does merit discussing some of the theoretical underpinnings of the approach. It will help us later in describing how it is effectively *de facto* an approach that functions at all levels of a design domain, as is required by systems thinking and engineering.

As mentioned, one of the fundamental theoretical precepts on which VSD is founded is that technology is not purely deterministic, instrumental, and/or socially constructed, but rather is *interactional*: it both supports and constrains social structures and vice versa. Consequently, technologies, and thus the social structures which they form and by which they are formed, can be described as *sociotechnical systems*. In the design of sociotechnical systems then, VSD is often described as a tripartite methodology constituted of: (1) conceptual investigations, (2) empirical investigations, and (3) technical investigations (see Fig. 1). They are often undertaken consecutively, in parallel, or iteratively. Respectively, they involve, (1) conceptual investigations into values and possible value tensions, 2) empirical investigations of the relevant stakeholders that enables one to define the way one understands their values and priorities, and determine their values, and 3) determination and evaluation of what the technical limitations of the technology and of the way they support and/or constrain identified values and design requirements.

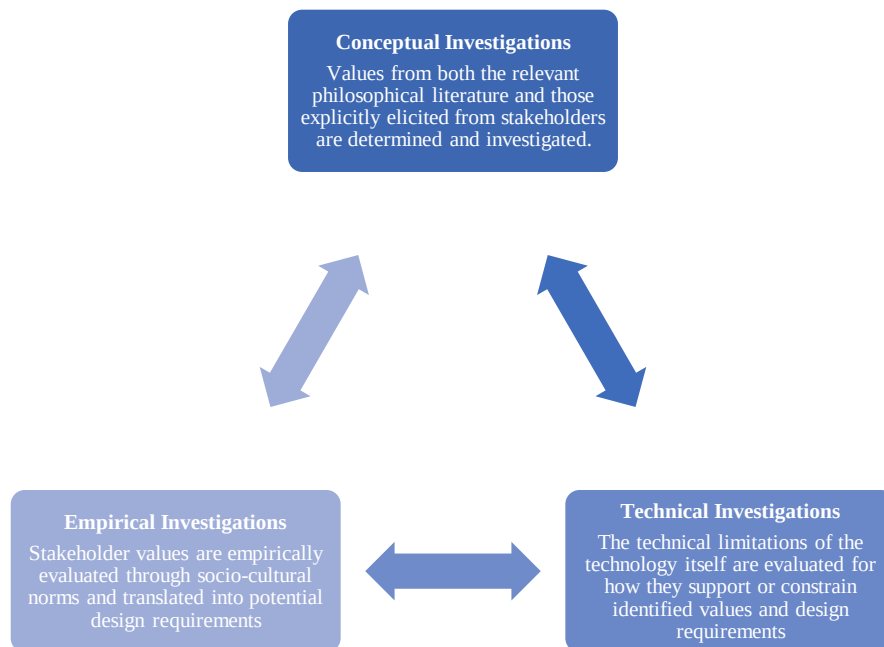


Figure 1. The recursive VSD tripartite framework (Source: Umbrello, 2020)

Although many VSD applications begin with conceptual investigations, it does not inflexibly specify as to which one to begin with. The approach is highly flexible and dependent on the

contextual particulars of any given domain. VSD projects can then be in one of three considerations: (1) the context of use, (2) a technology itself, and/or (3) a value(s) (fig. 2).

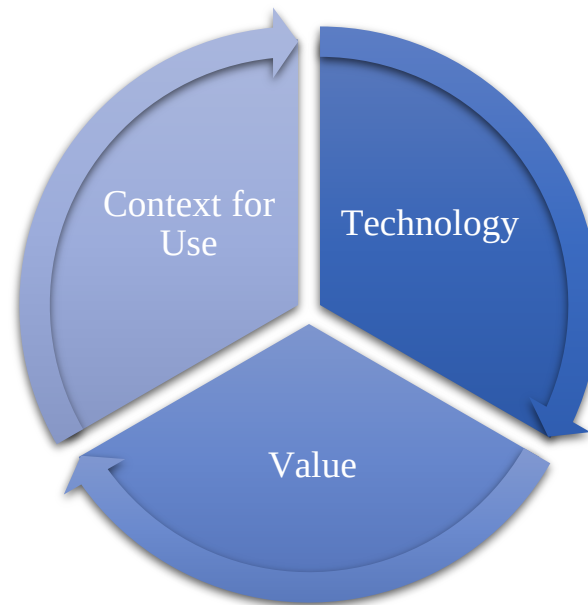


Figure 2. Starting considerations for VSD. Typically, one of the three is most pertinent to any given design. (Source: Umbrello, 2021)

Friedman and Hendry (2019) propose 17 more specific methods that can be used in VSD: (1) stakeholder analysis; (2) stakeholder tokens; (3) value source analysis; (4) coevolution of technology and social structure; (5) value scenarios; (6) value sketches; (7) value-oriented semi-structured interview; (8) scalable assessments of information dimensions; (9) value-oriented coding manual; (10) value-oriented mock-ups, prototypes, and field deployments; (11) ethnography focused on values and technology; (12) model for informed consent online; (13) value dams and flows; (14) value sensitive action-reflection model; (15) multi-lifespan timeline; (16) multi-lifespan co-design; and (17) envisioning cards. Each of the different methods that have been used over the three decades aim to address different points within the VSD investigations, and are not mutually exclusive. Nor can they be exhaustive given that novel approaches from the social sciences and other disciplines emerge as useful means of carrying out the three investigations.

In its goal towards designing technologies *for* human values, the methods aim at purposes such as identifying the direct and indirect stakeholders and stakeholder legitimation, expanding design spaces and identifying value sources, eliciting and representing values, analysing values in relation to social structure, as well as multi-generational envisioning and design thinking. Different VSD methods are better suited for different goals different technologies. However, we have to keep in mind that stakeholders are not only those outside the design domain, like public users, but also the designers themselves, management, and other members of design organizations. It's for this reason that VSD should be envisioned not only as a bottom-up approach adopted by designers, but also an approach supporting and constraining top-down structures in a design space, forming those structures and being informed by them. For this reason, VSD is not monolithic or confined to the design level proper, but are useful across the

various levels of technology design. Consequently, describing VSD as having multiple levels of abstraction is important not only for the ethical design of technologies, but also in ensuring that equifinality is obtained. The entire sociotechnical structure from which innovations are created are similarly constrained by the approach. The following section describes how VSD can be conceptualized across these levels of abstraction towards the goal of equifinality in both design and the design spaces.

VSD to design *for* Equifinality Via Agile

As already mentioned, VSD takes the sociotechnicity of technologies as a fundamental precept in its approach. However, if VSD aims to obtain true equifinality, that is, a symbiosis in which a particular system or technology forms a part, then divorcing the approach from different levels of the wider structure like managerial domains, and confining it primarily to the level of designers does a disservice. Up until this point, VSD has been applied and explored by designers themselves who have sought to explore its applicability and strengths. Its successes and methodological aptitude in designing *for* human values entitle it for larger-scale adoption. However, in order to promote it, we need to look at an instance that couples the various levels of abstraction, more specifically the designer levels and the managerial levels, and illustrate how VSD works in both directions (bottom-up and top-down). We can look at Agile as a vehicle for this integration of VSD. Agile practices call for collaborative cross-functional and self-organizing teams, relying on managerial levels to choose which work to be prioritised while designer levels organise around the tasks. This enables co-construction of the overall project, bringing in both bottom-up and top-down application of a central project vision. The result is a technology developed by all levels (i.e., *systems-within-systems*). Agile creates an internal organisational environment that is otherwise lacking in VSD and provides a landscape on which VSD can be modulated.

The tripartite methodology of VSD, which is comprised of conceptual, empirical, and technological investigations, naturally fit into the iterative cycle of planning, executing, and evaluating that is characteristic of an agile team (see Fig. 3). Conceptual investigations begin during planning, as values are explored and provisionally designed into a system, and ends during evaluation whereas as teams reflect on potential value tensions and consider how to account for them in the next iteration. In between planning and evaluation, the execution phase creates space for technical investigations, as a team envisages the building and implementation of the design. Finally, due to Agile's focus on stakeholder feedback at the end of each iteration, empirical investigations into stakeholder values and priorities obtain necessary insight in the evaluation phase.

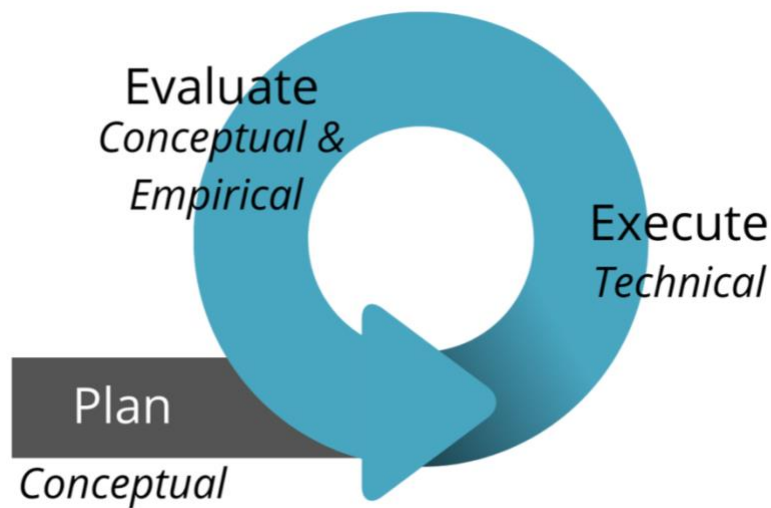


Figure 3. The mapping of the VSD investigations onto the Agile cycle.

Communication and adaptation is the core of Agile. Teams work within tight feedback cycles and strive for continuous improvement towards the central project vision. Agile has been criticised as a potential cause of poor decision making which results in an unethical technology due to its very quick pace and occasional lack of high-level impact assessment. However, that does not mean it is not a useful vehicle for value consideration, as Agile does create the structure by which VSD can be integrated to a project cross-functionally and multi-laterally throughout its lifecycle. In fact, the longer-term view and planning from VSD, if properly embraced in a team, has the potential to solve these concerns of the short-termism caused by Agile practices.

Implementing VSD in Engineering Teams: An Agile Approach

In order to further explore the prospect of utilising Agile as a means for implementing VSD, let us examine the application of one of the seventeen different VSD methods — the *Value Dams and Flows*—to the Agile cycle.

The *Value Dams and Flows* method focuses on avoiding system features which even a small number of stakeholders views as problematic. This involves identifying values as stakeholders want, and seeking a design which best reflects them, while continually addressing values-oriented trade-offs (Miller et al., 2007). A Value Dam refers to a technical feature or organisational policy a minority subgroup of stakeholders strongly oppose, which requires a serious consideration about its negative impact on them. Value Flows, on the other hand, are the values a large majority of the stakeholders would like to see embodied in the system, regardless of whether the resulting technical feature or organisational policy is necessary to the successful operational function of the system. The key is that once these dams and flows have been identified, designers must find a way to balance the two in order to address any values-oriented design trade-offs and/or moral overload of (van den Hoven et al., 2012).

At the end of every iteration of the Agile cycle, a new system or feature is deployed to stakeholders for feedback. This feedback is then used throughout the next planning phase to determine what feature will be built in the following iteration. This stakeholder feedback mechanism within the Agile cycle is an ideal vehicle for the *Value Dams and Flows* toolkit. It first creates the engagement with stakeholders necessary for identifying the various dams and flows, and then prioritises the insight gained from this feedback in designing the new features of a system.

Like most Agile workflows, they are broken down into ‘sprints’ or iterations, similar to how VSD iteratively designs innovations towards ever-greater equifinality. These pre-determined sprints allow for critical redesigning to take place as new information emerges across the workflow. Likewise, as figure 5 shows, redesigning as a function of mock-ups and prototypes is a key VSD method, just as it is inextricable from Agile. In doing so, what this permits is Agile teams to continue their current workflow patterns while simultaneously integrating a method of accomplishing it that is explicitly oriented to designing *for* human values.

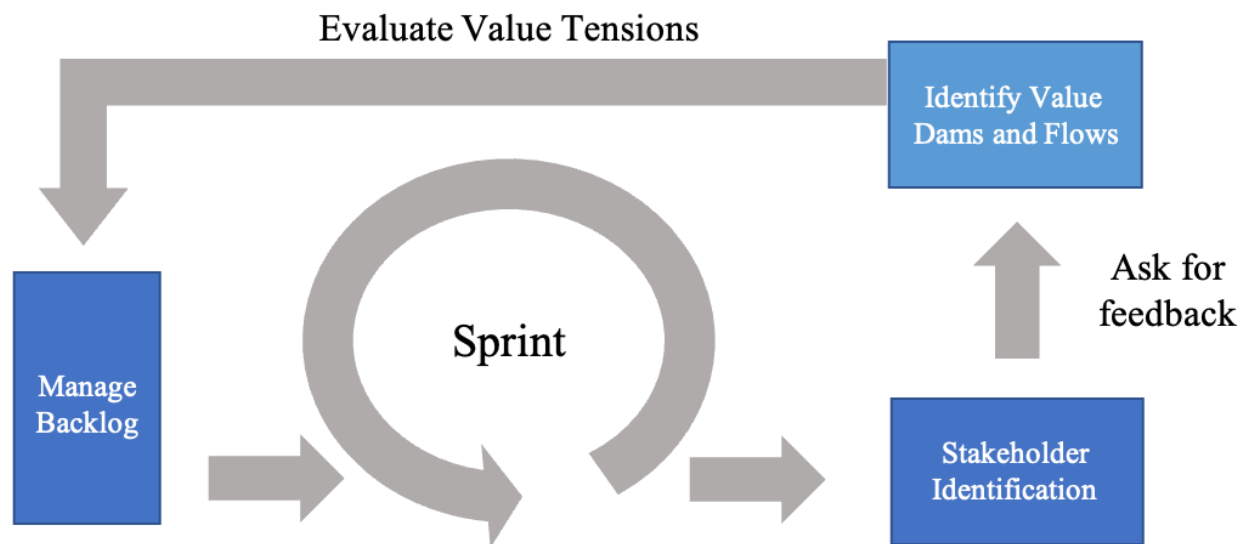


Figure 4. *Value Dams and Flows* Toolkit in the Agile cycle.

The above figure is by no means an exhaustive representation of the potential of VSD tools, which can be employed at any given stage of the Agile cycle. Multiple tools should be used across the Agile process given that the various stages of any given cycle parallels a tool in VSD that focuses on designing *for* human values, and can be mapped on to the cycle. For example, ‘deploy to stakeholders’ requires knowledge of *who* the relevant direct and indirect stakeholders are as well as *what* constitutes safe, yet salient deployment to those stakeholders. The former can be addressed by using two VSD tools: (1) *stakeholder analysis* for the identification and legitimization of the different relevant stakeholders (e.g., Czeskis et al., 2010) and (2) *stakeholder tokens* to facilitate the identification and articulate the interaction between stakeholders (Yoo, 2018). The latter regarding safe deployment can be resolved through *value-oriented mock-ups*, *prototypes*, or *field deployments* to determine if the technology safely embodies the design values and to evaluate the system for potential recalcitrance, albeit in an incremental and controlled environment. By observing for any previously unforeseen (or unforeseeable) unwanted values, this can trigger another design iteration (i.e., redesign) [see Fig. 5].

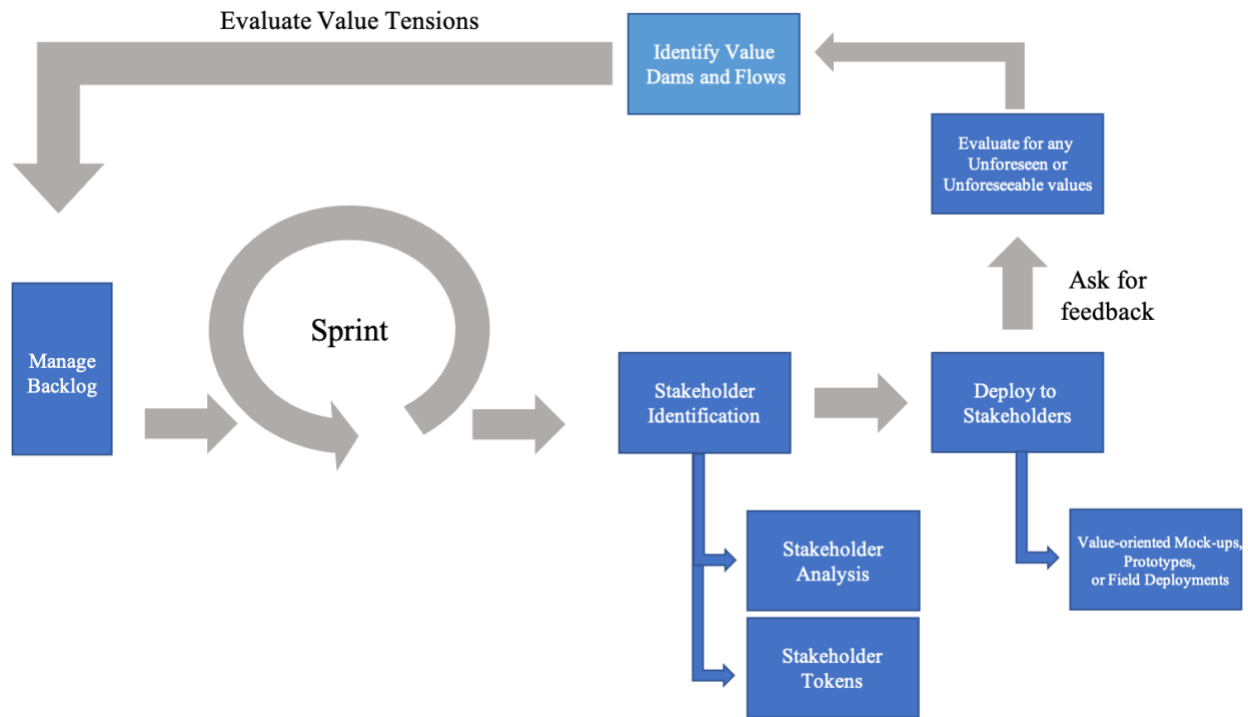


Figure 5. Agile workflow with the addition of VSD stakeholder identification tools

In ‘evaluating value trade offs’, or ‘moral overload’ as it is referred to in VSD scholarship there are existing tools to help designers to manage this stage of development (van den Hoven et al., 2012). The language of ‘trade-offs’, which is often framed in cost-benefit analysis language is exchanged in VSD, and this paper, for a more inclusive approach to value tensions by co-creating value representations with stakeholders to determine points of overload, and how to design systems to support and constrain such tensions via compromise. *Value scenarios, value sketches, and the value sensitive action-reflection model* are all potential tools that Agile team members can employ to establish a connection with the value understandings, and consequently the technical design requirements for the values at play. Figure 6 illustrates in greater detail the process for how Agile can use VSD. It’s important to note the symbiotic relationship that VSD and Agile have when they are paired together; VSD is not overlaid on top of Agile, and neither does Agile becomes overly complicated with the addition of VSD. In fact, Agile already possesses many of the underlying ontological tools that allow an approach like VSD to integrate seamlessly into its workflow.

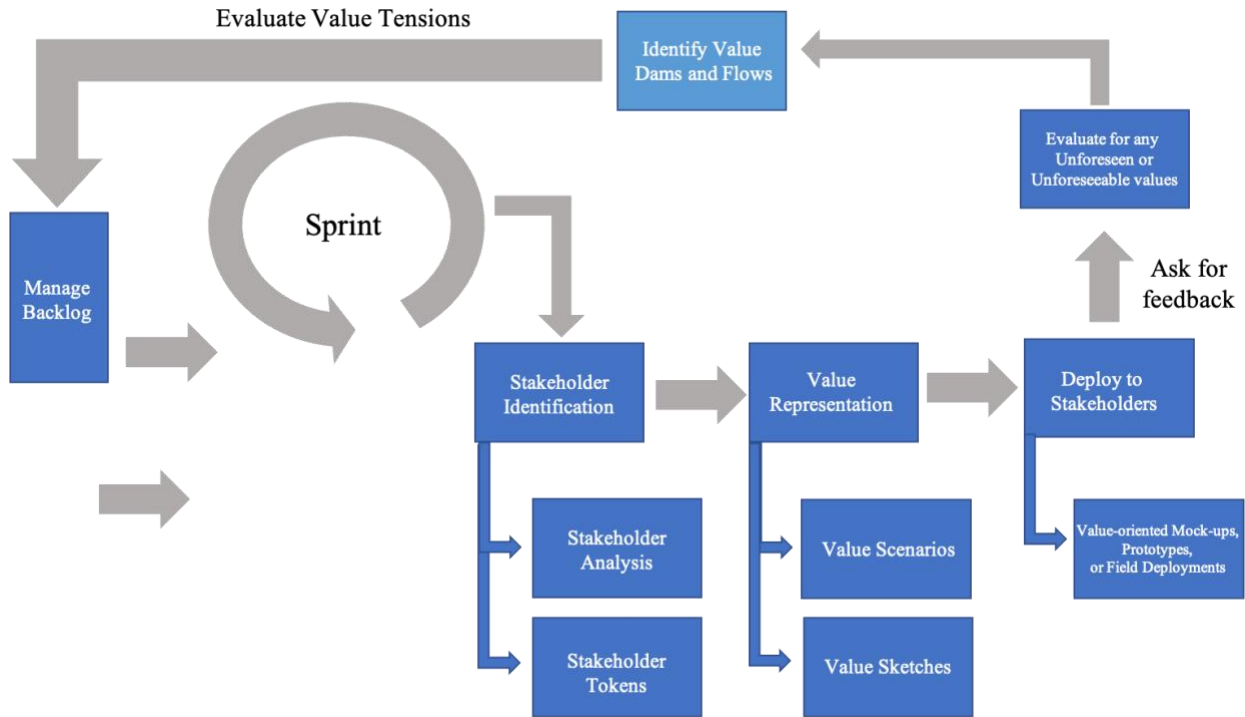


Figure 6. Agile workflow with the addition of VSD value representation tools

Umbrello and van de Poel (2021) argue that the VSD tool of ‘value hierarchies’ can be employed for artificial intelligence systems to help designers to translate different values (i.e. higher-order values, AI specific values, and stakeholder/contextual values) through norms, and then into design requirements (and vice versa). Similar heuristics can be used to streamline the evaluation of these apparent ‘trade off’ scenarios by Agile design teams (e.g., Umbrello, 2018; van de Kaa et al., 2019; van de Poel, 2013, 2014, 2017; van den Hoven et al., 2012). Nonetheless, what we have illustrated here is that implementing a design-for-values approach, like VSD, is achievable not via wholesale overhauls of design domains and programmes, but by invoking one of the fundamental tenets of VSD, i.e., that VSD can and should be integrated into existing practices and workflows. Here, we chose to use Agile because it is a popular, globally-adopted workflow used by design teams. However, given that Agile is not fundamentally devised towards designing *for* human values, and in view of the growing importance of this in the modern world, we offer VSD as a toolkit that can be seamlessly integrated into current Agile workflows. A key benefit of the adoptability of VSD is that the degree to which its use can be augmented as needed to any given situation, while still promoting the benefits of designing *for* human values.

Conclusions

Value sensitive design (VSD) is a principled approach to design that is increasingly being adopted for a wide range of technologies, and across many domains of use, given its potential for designing technologies *for* human values. To date, however, the literature on VSD focuses almost exclusively on the technologies themselves, or on the nature and state of human values in the methodological framework. Because VSD is predicated on the concept that technologies are

sociotechnical, social structures influence and are being influenced by technology. This merits an examination of the social structures that support and constrain the domains themselves, e.g., managerial levels of design domains. This chapter argues that in order for VSD to achieve a truly salient design *for* human values, its tenets of support and constraint must operate seamlessly across all workflow levels, from the design domain to the domain of management. We have shown how team members at the managerial levels who employ project management methodologies like Agile can adopt VSD as a toolkit, and how VSD can be used to iteratively redesign the internal roles and policies of the design domain towards a greater equifinality of innovation.

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